Opinion Network Formation

Assignment 1: LinkedOut

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1 Team Name

LinkedOut

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2 GitHub Link to Code

Repository: https://github.com/dawnbliss-coder/opinion-network-formation

3 Dataset Documentation

3.1 Data Description

3.2 Encoding and Preprocessing

3.3 Handling Missing Data and Data Quality

4 Pipeline Followed

4.1 User–User Network

4.2 Statement–Statement Network

4.3 User–Statement Network

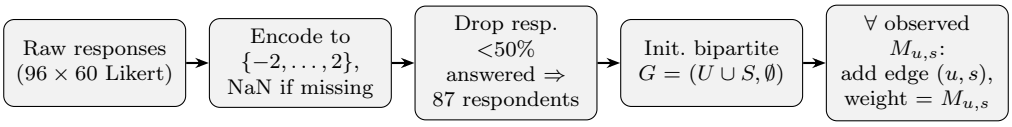


Figure 1: User–Statement network construction pipeline.

Formally, let $M \in \{-2, -1, 0, 1, 2, \text{NaN}\}^{U \times S}$ be the encoded, engagement-filtered response matrix ($|U|$=87 respondents, $|S|$=60 statements). The network is the bipartite graph $G = (U \cup S, E)$ with

$(u, s) \in E \iff M_{u,s}$ is observed (not NaN), and edge weight $w_{u,s} = M_{u,s} \in \{-2, -1, 0, 1, 2\}$. No edge is created for a skipped/"No Comments" entry, since skipping $\neq$ a Neutral (0) response.

A reference check (`edge_threshold_table()` in `src/user_statement_network.py`) confirms that sparsifying by response magnitude does not produce a meaningfully sparser graph:

Cutoff	Edges	Density
All answered (incl. Neutral)	5,158	98.8%
Drop Neutral only	4,491	86.0%
Strongly Agree/Disagree only	2,349	45.0%

5 Analysis and Visualizations

5.1 User–User Network

5.2 Statement–Statement Network

5.3 User–Statement Network

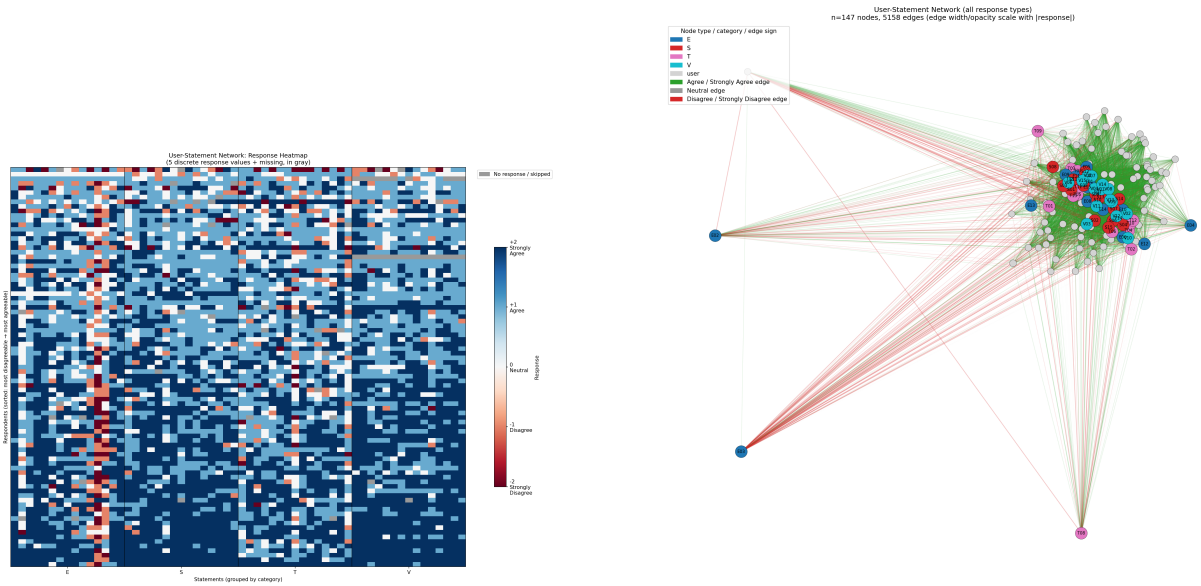


Figure 2: *Left:* response heatmap (respondents sorted most disagreeable → most agreeable; statements grouped by category; 5 response colors + gray for missing). *Right:* full bipartite network, all 5,158 edges and all five response types drawn (green = agree, red = disagree, gray = neutral; width/opacity $\propto$ |weight|).

Metric	Value
Users / Statements	87 / 60
Edges (answered)	5,158 / 5,220 possible
Density $= \lvert E\rvert/(\lvert U\rvert\lvert S\rvert)$	98.8%
Mean edge weight $= \frac{1}{\lvert E\rvert}\sum_{(u,s)\in E} w_{u,s}$	+1.14
Response split	80.0% positive, 12.9% neutral, 7.1% negative

Density near 100% means missingness does not carry much structure here. Almost everyone answered almost everything, so the real signal lives in edge *weight* rather than edge presence. The positive mean weight shows that the class leans toward agreement across the survey as a whole.

Per-node metrics: let $N(u) = \{s : (u, s) \in E\}$ (statements u answered) and $N(s) = \{u : (u, s) \in E\}$ (respondents who answered s), so $\deg(u) = |N(u)|$, $\deg(s) = |N(s)|$.

Metric	Formula	What it tells us
mean_response(u)	$\frac{1}{\lvert N(u)\rvert}\sum_{s\in N(u)} w_{u,s}$	Respondent u's average stance. Used to rank who is most agreeable and who is most disagreeable overall.
mean_abs_response(u)	$\frac{1}{\lvert N(u)\rvert}\sum_{s\in N(u)} \lvert w_{u,s}\rvert$	Respondent u's opinion *intensity*, independent of direction. This separates decisive respondents from wishy-washy ones (someone alternating between Strongly Agree and Strongly Disagree scores high here even if their mean_response is close to 0).
mean_response(s)	$\frac{1}{\lvert N(s)\rvert}\sum_{u\in N(s)} w_{u,s}$	Statement s's consensus level and direction (near ± 2 means broad agreement or disagreement, near 0 means the respondents are split).
std_response(s)	$\sqrt{\frac{1}{\lvert N(s)\rvert - 1}\sum_{u\in N(s)} (w_{u,s} - \bar{w}_s)^2}$	Spread of opinion on s. A low std together with a high |mean| points to a consensus statement, while a high std points to a polarizing or contested statement, regardless of its average.

By these metrics, consensus statements (high mean, low std) include E15 and V13, while the most polarizing statements (high std) are E02, E03, and E04. It's worth noting that all three of the polarizing ones come from the Education category.

6 Results and Discussion

6.1 User–User Network

6.2 Statement–Statement Network

6.3 User–Statement Network

The near-total density (98.8%) combined with a positive mean edge weight (+1.14) shows the class leans broadly toward agreement across the survey, consistent with the 80%/13%/7% positive/neutral/negative split (Section 5.3): most respondents answered most statements, and most answers were Agree or Strongly Agree.

At the respondent level, mean_response isolates a small set of genuine outliers from the agreeable majority. Respondent 120 has the lowest mean_response (-0.21), the only respondent whose average stance falls on the disagreement side of neutral despite answering 56 of 60 statements. That looks like a real dissenting voice rather than a low-engagement artifact.

At the statement level, std_response identifies the Education-category statements E02, E03, and E04 as the most contested in the entire survey, while E15 and V13 are the strongest consensus statements. It's worth pointing out that **S09**, the most structurally central statement in the Statement–Statement network (highest degree and eigenvector centrality), also shows up among the most consensus statements here (mean 1.47, std 0.59). So S09's centrality seems to come from *broad shared agreement* with many other statements rather than from controversy: centrality and polarization are measuring two different things.

6.4 Cross-Network Synthesis

7 Individual Contribution

Member	**Contribution**
Name 1	
Name 2	
Priyanka Agarwal	User–Statement network: bipartite graph construction (`user_statement_network.py`), response-weighted edge scheme, per-user/per-statement metrics, heatmap and full-network visualizations, corresponding report sections.